UNIVERSIDAD TORCUATO DI TELLA

TD7: Ingeniería de Datos

Big Data. Motivación

Big Data

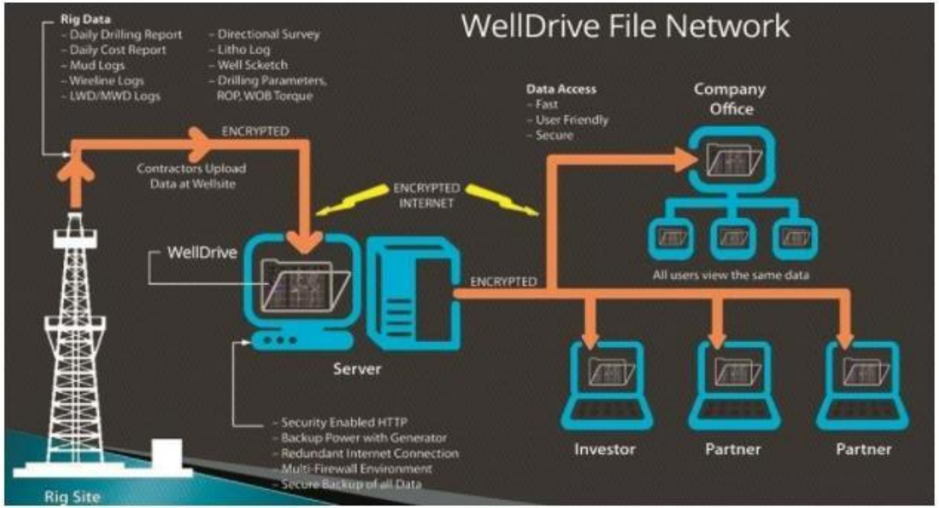
Big Data

Principles and best practices of scalable real-time data systems

Nathan Marz
James Warren

MANNING

Las cuatro “V” del Big Data: volumen



“Research has determined that **a single well can produce about one terabyte of production data - every day.** That must be processed, organized and stored by someone, and can take hours of company time to do so.”

[Fuente: Production and operational data management made easier |Oil & Gas Product News (oilandgasproductnews.com)]

Las cuatro “V” del Big Data: volumen

London Stock Exchange completes acquisition of Refinitiv

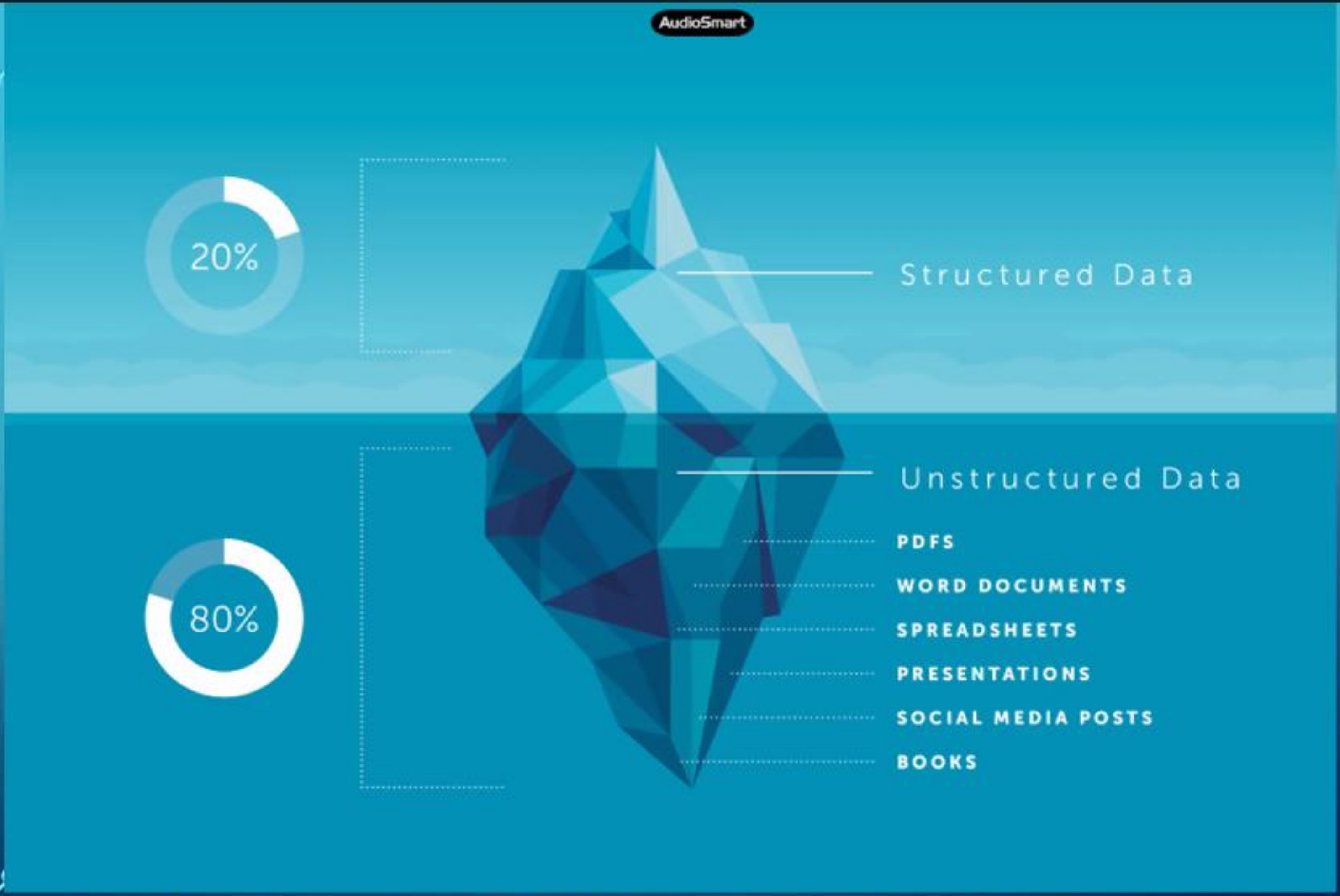
29 January 2021 1 3 1

Source: London Stock Exchange

Further to the announcement on 28 January 2021, London Stock Exchange Group plc (“LSEG” or the “Company”) confirms that its all-share acquisition of Refinitiv (the “Transaction”) has now completed.

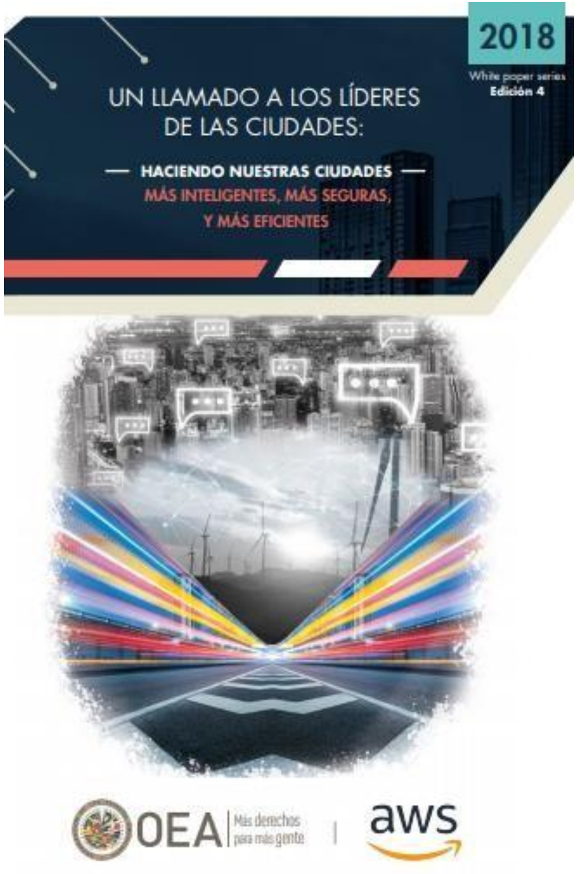
“It collects, aggregates, analyses, curates, stores and distributes huge volumes of data from a large number of sources. It collects more than one petabyte of data every day and processes more than a million unstructured documents. Its data platform provides 40 billion market updates every day.”

Fuente: https://www.computerweekly.com/news/252467734/London-Stock-Exchange-agrees-to-acquire-Refinitiv-for-22bn



https://lawtomated.com/structured-data-vs-unstructured-data-what-are-they-and-

Las cuatro “V” del Big Data: variedad

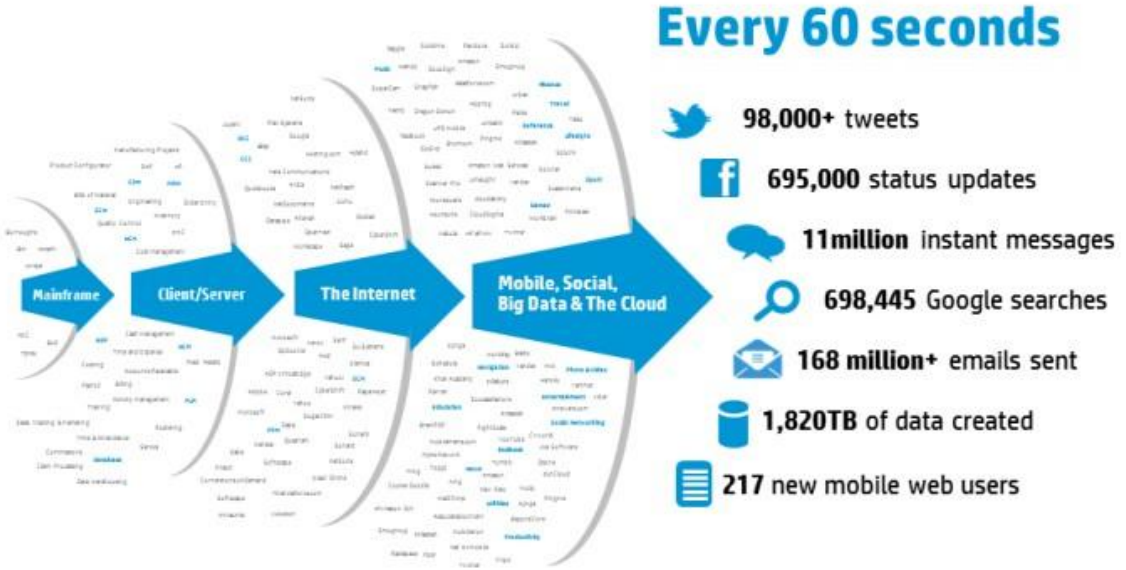


- Base tecnológica de una ciudad inteligente, compuesta por sensores, comunicaciones y portales de datos abiertos.
- Procesamiento geográfico combinado con información de tráfico.
- Videos procesados en tiempo real.
- Sensores lumínicos, de energía, de concentración de NO2.

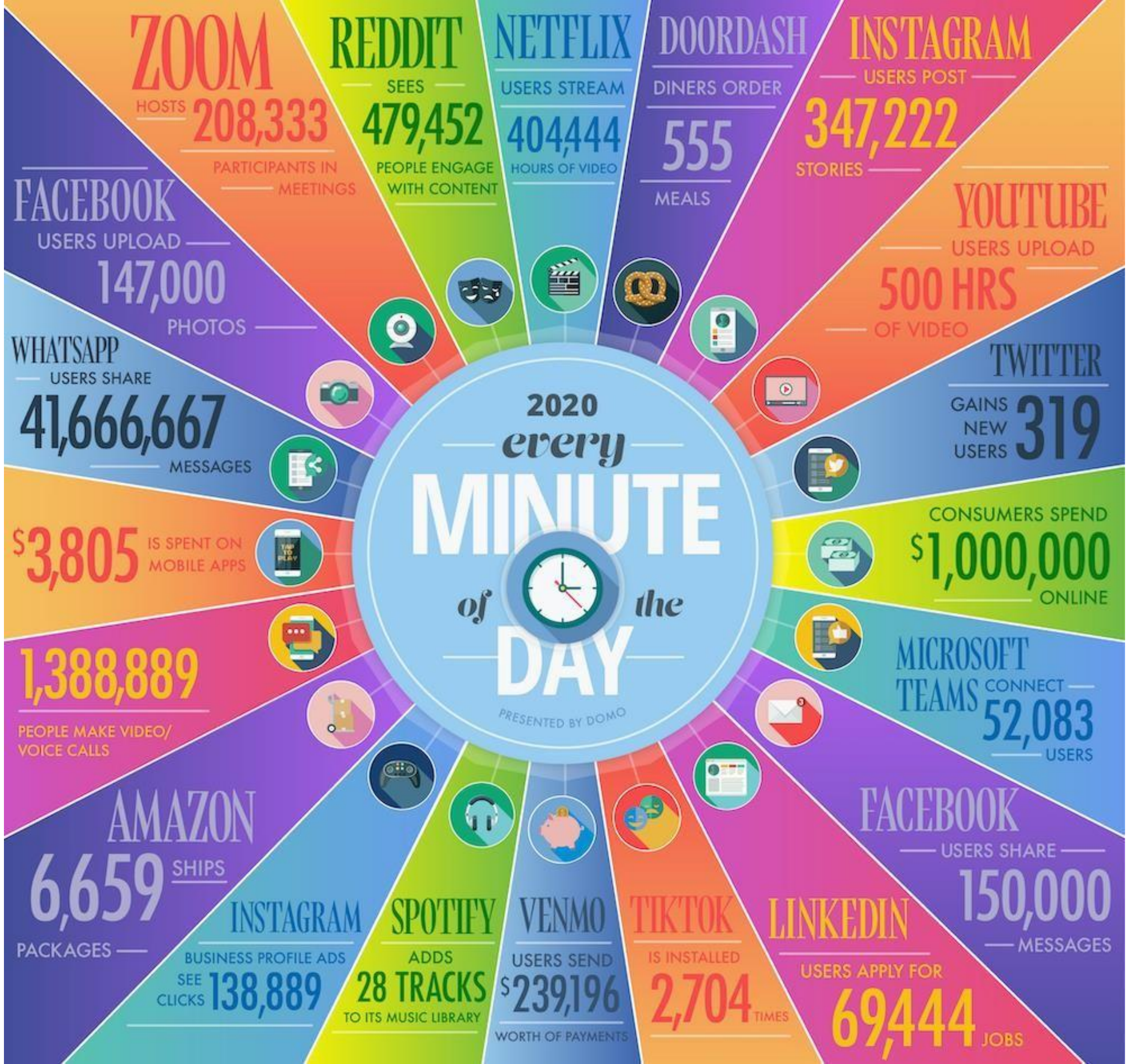
Fuente: http://www.oas.org/es/sms/cicte/ciberciudades.pdf

Las cuatro "V" del Big Data: variedad (tipos de datos nuevos)

IDC predicts that by 2015 over 90 percent of that data will be unstructured (e.g., images, videos, MP3 music files, and other files based on social media and Web enabled workloads). While this data is full of rich information it is hard to understand and analyze[2]. For many organizations, they haven't even begun to understand how to do it with these new data types.



Fuente:
https://www.hp.com/hpinfo/newsroom/press_kits/2012/HPDiscoverFrankfurt2012/IO_Whitepaper_Harness_the_Power_of_Big_Data.pdf



Fuente:
Data
Never
Sleeps 8.0

domo.com

Las cuatro “V” del Big Data: velocidad

THE WEEK

OPINION BRIEFINGS SPEED READS TALKING POINTS CARTOONS MORE

IN-DEPTH BRIEFING

Wall Street's secret advantage: High-speed trading

They're unknown and invisible to most of us, but electronic trading programs now rule the stock markets

THE WEEK STAFF
JANUARY 11, 2015

What is high-speed trading? It's Wall Street's winning edge. By harnessing massive computer power to buy and sell stocks in the blink of an eye, high speed traders leverage tiny changes in value to make huge profits. The technique was pioneered in the

“**Does co-location increase speed?** Yes. The physical proximity to the exchange server reduces the time from when a firm’s buy or sell order is entered and when it’s executed. “By co-locating,” says Adam Afshar of Hyde Park Global, a high-speed trading firm, “we are able to take 21 milliseconds off our trades. In the past, 21 milliseconds was a trivial matter. Now it’s a pivotal matter.” Several academic studies have found that shaving even one millisecond off every trade can be worth $100 million a year to a large, high-speed trading firm.”

Fuente: https://theweek.com/articles/493238/wall-streets-secret-advantage-highspeed-trading

Las cuatro "V" del Big Data: veracidad

Semantic Web - Interoperability, Usability, Applicability *an IOS Press Journal*

Linked Data, Big Data, and the 4th Paradigm

Editorial

Pascal Hitzler, [a] Krzysztof Janowicz, [b]
[a] *Kno.e.sis Center, Wright State University, USA*
[b] *University of California, Santa Barbara, USA*

"Big Data, an increasing number of V's has been used to characterize different dimensions and challenges of Big Data: volume, velocity, variety, value, and veracity. Interestingly, different (scientific) disciplines highlight certain dimensions and neglect others. For instance, super computing seems to be mostly interested in the volume dimension while researchers working on sensor webs and the internet of things seem to push on the velocity front. The social sciences and humanities, in contrast, are more interested in value and veracity."

Fuente: http://www.semantic-web-journal.net/system/files/swj488.pdf

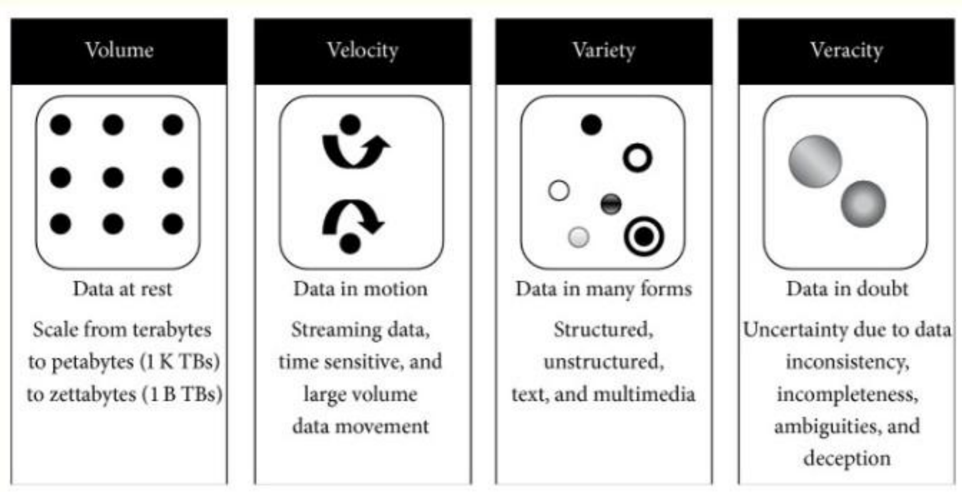
Las cuatro "V" del Big Data: veracidad

The Scientific World Journal

ScientificWorldJournal. 2015; 2015: 453597. PMCID: P
Published online 2015 Oct 1. PMI
doi: 10.1155/2015/453597

Distilling Big Data: Refining Quality Information in the Era of Yottabytes

The data quality can generally be grouped under several categories like accuracy, believability, reputation, objectivity, factuality, consistency, freedom from bias, correctness, and unambiguousness.



Fuente: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4606078/#

Big Data

En resumen, se estima que en 2025, cada día se produjeron **400 millones de terabytes** de datos (ó 0,4 zettabytes diarios)

Esto implica **4600 terabytes** por **segundo**

¿Dónde se genera tanta información?

¿De qué formas puede ser explotada?

Big Data

¿Cómo procesar este tipo de datos?

- Online vs Offline
- Batch vs Real-time
- Cómputo Secuencial vs Paralelo
- Estructurado vs No Estructurado
- Sistemas distribuidos vs Centralizados

BLOG@CACM

The End of a DBMS Era (Might Be Upon Us)

By Michael Stonebraker
June 30, 2009
Comments (7)

VIEW AS: SHARE:

Relational database management systems (DBMSs) have been remarkably successful in capturing the DBMS marketplace. To a first approximation they are "the only game in town," and the major vendors (IBM, Oracle, and Microsoft) enjoy an overwhelming market share. They are selling "one size fits all"; i.e., a single relational engine appropriate for all DBMS needs. Moreover, the code line from all of the major vendors is quite elderly, in all cases dating from the 1980s. Hence, the major vendors sell software that is a quarter century old, and has been extended and morphed to meet today's needs. In my opinion, these legacy systems are at the end of their useful life. They deserve to be sent to the "home for tired software."

Fuente: https://cacm.acm.org/blogs/blog-cacm/32212-the-end-of-a-dbms-era-might-be-upon-us/fulltext